

Example 1 – Diode Load-Line Analysis

Analyze a diode circuit using load-line analysis to determine the operating point (Q-point), where the diode's current and voltage satisfy both the circuit constraints and diode I-V characteristics.

Power Supply Voltage: $V_{SS} = 5V$

Series Resistor: $R = 1 \text{ k}\Omega$

Diode Parameters:

Saturation Current: $I_S = 10^{-12} \text{ A}$

Thermal Voltage: $V_T = 25 \text{ mV}$

Diode I-V Relationship:

$$I_D = I_S \left(e^{\frac{V_D}{V_T}} - 1 \right)$$

$$V_{SS} = I_D R + V_D$$

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$$I_D = \frac{V_{SS} - V_D}{R} = \frac{5 - V_D}{1000}$$



This equation represents a straight line in the I-V plane, known as the load line.

Example 1 – Diode Load-Line Analysis

The diode's current-voltage relationship is modeled by:

$$I_D = 10^{-12} (e^{V_D/25mV} - 1)$$

This equation defines the nonlinear behavior of the diode.

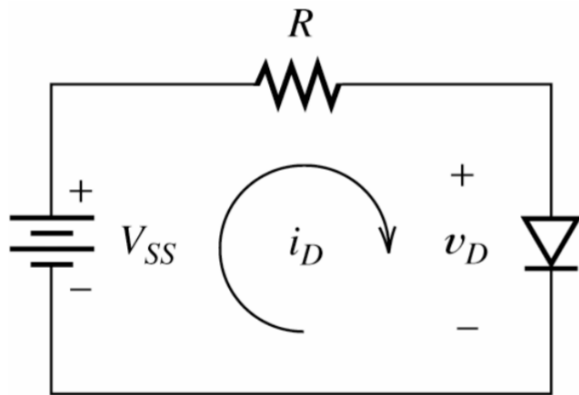
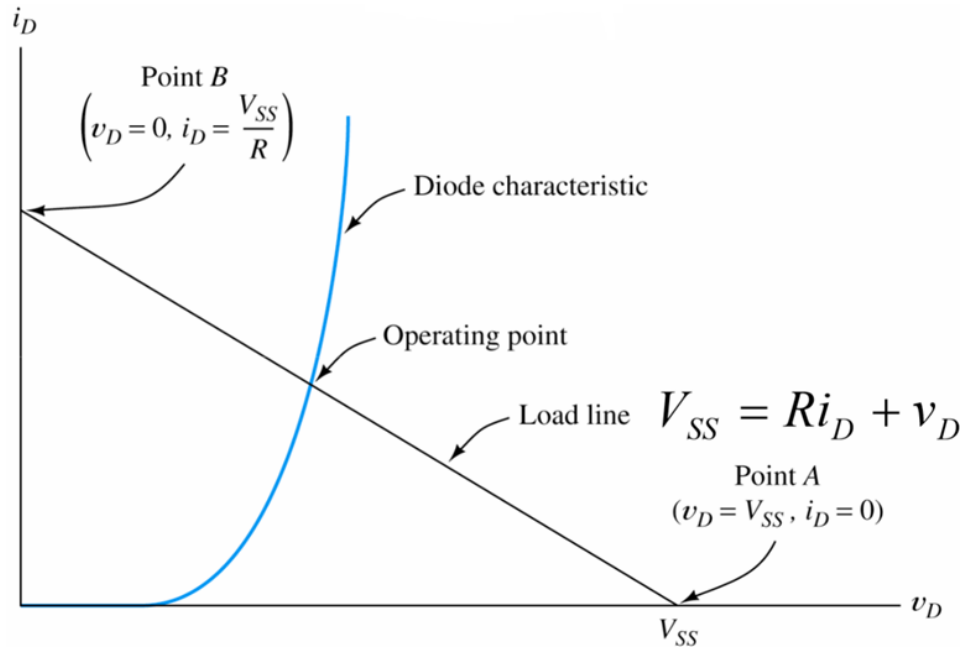
Finding the Q-Point (Operating Point):

The Q-point (operating point) is determined by finding the intersection of the load line and the diode I-V curve. This requires solving:

$$\frac{5 - V_D}{1000} = 10^{-12} (e^{V_D/25mV} - 1)$$

Since this equation is nonlinear, it must be solved using graphical or numerical methods.

Example 1 – Diode Load-Line Analysis



$$I_D = \frac{V_{SS} - V_D}{R} = \frac{5 - V_D}{1000}$$

When $V_D = 0$ V, then $I_D = 5$ V / 1 k Ω = 5 mA

When $V_D = 5$ V, then $I_D = 0$ A.

$V_D \approx 0.7$ V assumption !!!

$$I_D = \frac{5 - 0.7}{1000} = 4.3 \text{ mA}$$

The approximate Q-point is:

$$V_D \approx 0.7 \text{ V}, I_D \approx 4.3 \text{ mA}$$

Example 2 – Design of a Common-Source Amplifier

Design a common-source MOSFET amplifier for an audio application. The amplifier should meet the following specifications:

1. Transconductance (g_m): At least 5 mS
2. Voltage Gain (A_v): At least 30 (29.54 dB)
3. Gate-Source Voltage (V_{GS}): Should be between 1 V and 2 V
4. Maximum Output Amplitude Peak-to-Peak, $V_{out, pp}$ V_{-} (Before Clipping): At least 8 V
5. Drain Resistor (R_D): 4.7 k Ω
6. Power Supply Voltage (V_{DD}): 12 V
7. MOSFET Parameters:
 1. Electron Mobility (μ_n) = 350 cm²/V·s
 2. Oxide Capacitance (C_{ox}) = 3.45×10^{-7} F/cm²
 3. Threshold Voltage (V_{th}) = 0.8 V
 4. Channel Length (L) = 0.8 μ m
 5. Drain Current (I_D): To be determined
 6. Channel Width (W): To be determined

Example 2 – Design of a Common-Source Amplifier

Fundamental equations:

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

Determine the drain current I_D needed to satisfy both g_m and V_{GS} constraints.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

Verify the voltage gain (A_v) and ensure it meets the required minimum.

$$A_v = -g_m R_D$$

Calculate the maximum allowable output amplitude to ensure the amplifier does not clip.

$$V_{DD} = V_{DSQ} + I_{DQ} R_D$$



If V_{DSQ} selected as 8 V ($V_{DSQ} = 8\text{ V}$)



$$12 = 8 + I_D \cdot 4700$$



$$I_D = 0.85\text{ mA}$$

Example 2 – Design of a Common-Source Amplifier

We first calculate g_m using

$$A_v = -g_m R_D$$

Rearrange for g_m :

$$g_m = \frac{|A_v|}{R_D}$$

Use $A_v = 30$ and $R_D = 4.7 \text{ k}\Omega$.

$$g_m = \frac{30}{4.7 \cdot 10^3} = 6.38 \text{ mS}$$

Drain Current I_D Calculation

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

and $W = 160 \text{ }\mu\text{m}$ (W is selected such that g_m is approximately equal to 6.38 mS)



$$g_m = 6.41 \text{ mS}$$

Example 2 – Design of a Common-Source Amplifier

Gate-Source Voltage V_{GS} Calculation

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad \Rightarrow \quad V_{GS} = V_{th} + \sqrt{\frac{2I_D}{\mu_n C_{ox} \frac{W}{L}}} \quad \Rightarrow \quad V_{GS} = 0.8 + \sqrt{\frac{2 \cdot 0.135 \cdot 10^{-3}}{350 \cdot 3.45 \cdot 10^{-7} \frac{10^{-3}}{0.8 \cdot 10^{-4}}}}$$

$$V_{GS} = 1.07 \text{ V}$$

Since $1\text{ V} \leq V_{GS} \leq 2 \text{ V}$, the condition is satisfied.

Example 2 – Design of a Common-Source Amplifier

Maximum Peak-to-Peak Output Amplitude $V_{out, pp}$

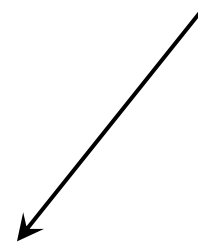
$$V_{DS, sat} = V_{GS} - V_{th} = 1.07 - 0.8 = 0.27 V$$



$$V_{outmax,p1} = (V_{DSQ} - V_{DS, sat}) = (8 - 0.27) = 7.73 V$$

$$V_{outmax,p2} = (V_{DD} - V_{DSQ}) = (12 - 8) = 4 V$$

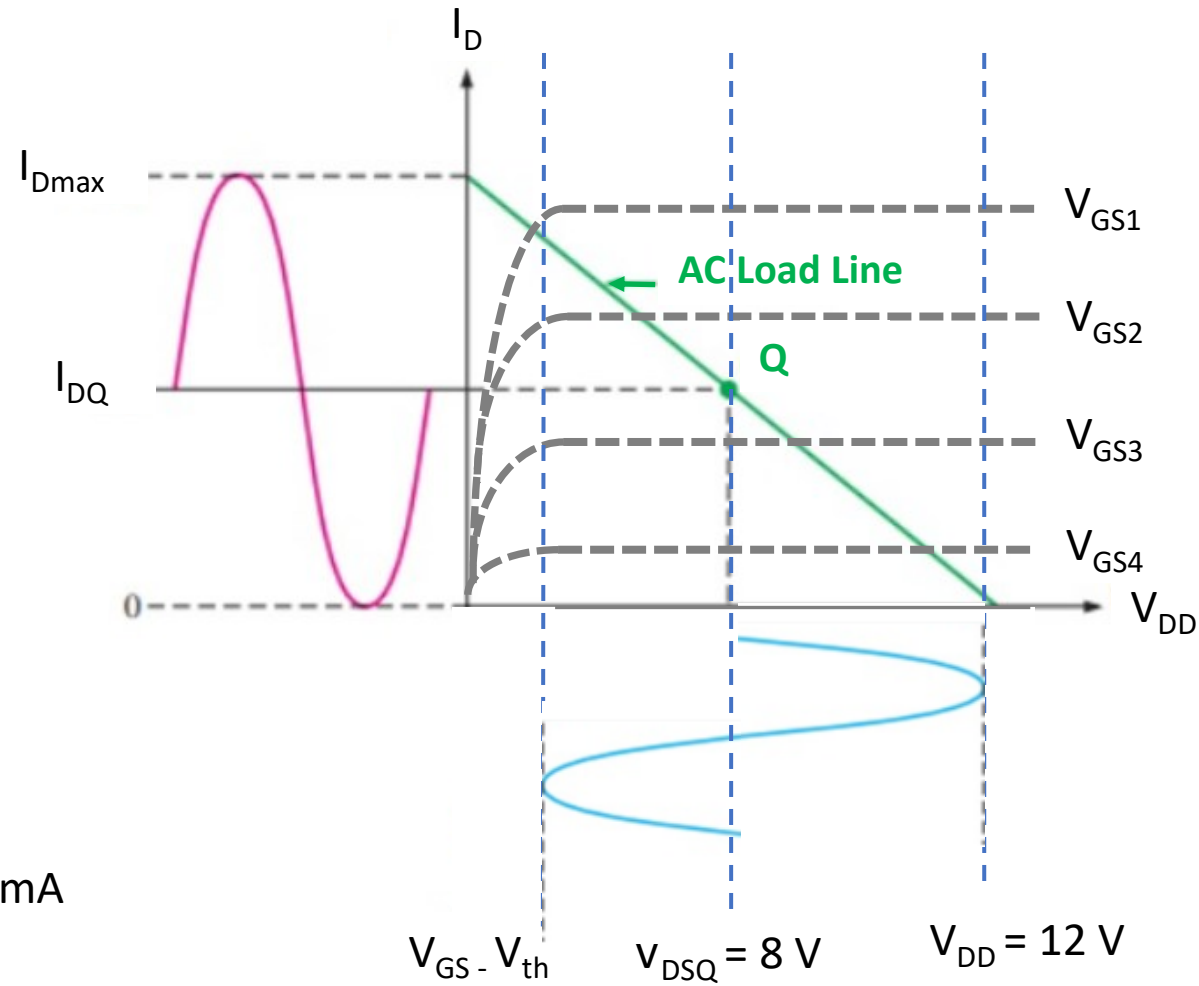
$$V_{outmax,pp} = 2 \cdot 4 = 8 V$$



The maximum V_{pp} is OK!

OK

Example 2 – Design of a Common-Source Amplifier



$$I_{dmax} = 12 / 4700 = 2.55\text{ mA}$$

$$12 = 4 + I_D \cdot 4700 \rightarrow I_D = 1.7\text{ mA}$$

$$12 = 8 + I_D \cdot 4700 \rightarrow I_{DQ} = 0.85\text{ mA}$$

Example 3 – BJT Amplifiers

Design a common-emitter BJT amplifier. Determine the operating point (Q-point) and verify that the transistor is in the active region.

Base Resistor: $R_B = 100 \text{ k}\Omega$

Supply Voltage: $V_{CC} = 12 \text{ V}$

Collector Resistor: $R_C = 500 \text{ }\Omega$

Emitter Resistor: $R_E = 100 \text{ }\Omega$

DC Current Gain: $\beta = 100$

Base-Emitter Voltage: $V_{BE} = 0.7 \text{ V}$

Example 3 – BJT Amplifiers

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta R_E)} = \frac{12 - 0.7}{100000 + (100 \cdot 100)} = \frac{11.3}{200000} = 102.7 \mu A$$

$$I_C = \beta I_B = 100 \cdot 0.057 \text{ mA} = 10.27 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 12 - (10.27 \text{ mA} \cdot (500 \Omega + 100 \Omega)) = 12 - (10.27 \cdot 0.6) = 5.838 \text{ V}$$

What does this mean? The transistor is in Active Region (Assumption: $I_C \approx I_E$, since I_B is too small)!

Condition for Active Region: $V_{CE} > V_{BE}$